STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 3: Motion in a Plane

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

Vector.

Q2 - 1 mark

Acostheta.

Q3 - 1 mark

Zero.

Q4 - 1 mark

Inward.

Q5 - 1 mark

5.

Q6 - 1 mark

v=romega.

Q7 - 1 mark

45 deg .

Q8 - 1 mark

Zero.

Q9 - 2 marks

A_x=5 N; A=8.66 N.

Q10 - 2 marks

Zero.

Q11 - 2 marks

24 m/s^2.

Q12 - 2 marks

2 s.

Q13 - 2 marks

10.

Q14 - 2 marks

4pi rad/s.

Q15 - 3 marks

T=2 s; H=5 m.

Q16 - 3 marks

10 N at 36.9 deg north of east.

Q17 - 3 marks

4 m/s^2 inward.

Q18 - 3 marks

3 s; 60 m.

Q19 - 4 marks

2 s; 5 m; 34.6 m.

Q20 - 4 marks

40 N inward; centripetal force does zero work.